

In-Lab exercise for polynomial (linear) fitting

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Goals

Today, we are going to look at how working out might affect one's life. We are going to look at some simulated (and not necessarily, but possibly realistic) data looking at how the number of hours per day somebody spends working out is related to:

- how “in shape” people report they feel
- resting heart rate (lower is considered more healthy)
- productivity, as assessed by a boss or supervisor

If we think about this for a hot minute, it seems reasonable that as people work out more:

- they probably feel more in shape
- their resting heart rate probably decreases, but only up to a point
 - initially, heart rate will decrease from unhealthy to healthy levels
 - once a healthy level is reached, heart rate can't drop (much) more
- their boss has mixed feelings
 - a little working out probably boosts productivity along with health
 - a LOT of working out might be viewed as a distraction (“Why is so-and-so at the gym when everybody else is here putting together the big project?”)

There are three data sets. For each, the first variable is average hours per day spent working out. The second variable is, for each data set in turn:

- a self-report of how “in shape” each person feels on a scale of 0 to 10
- average resting heart rate in beats per minute (bpm)
- a “boss-report” productivity rating on a scale of 0 to 10

The data sets required are in the files:

- shapeVsHours.csv
- bpmVsHours.csv
- prodVsHours.csv

Tutorial: Reported shape and heart rate datasets

1. Load shapeVsHours.csv and take a look

First things first: load the tidyverse

```
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.2
```

```
## v ggplot2 4.0.0      v tibble 3.3.0
## v lubridate 1.9.4    v tidyr 1.3.1
## v purrr 1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

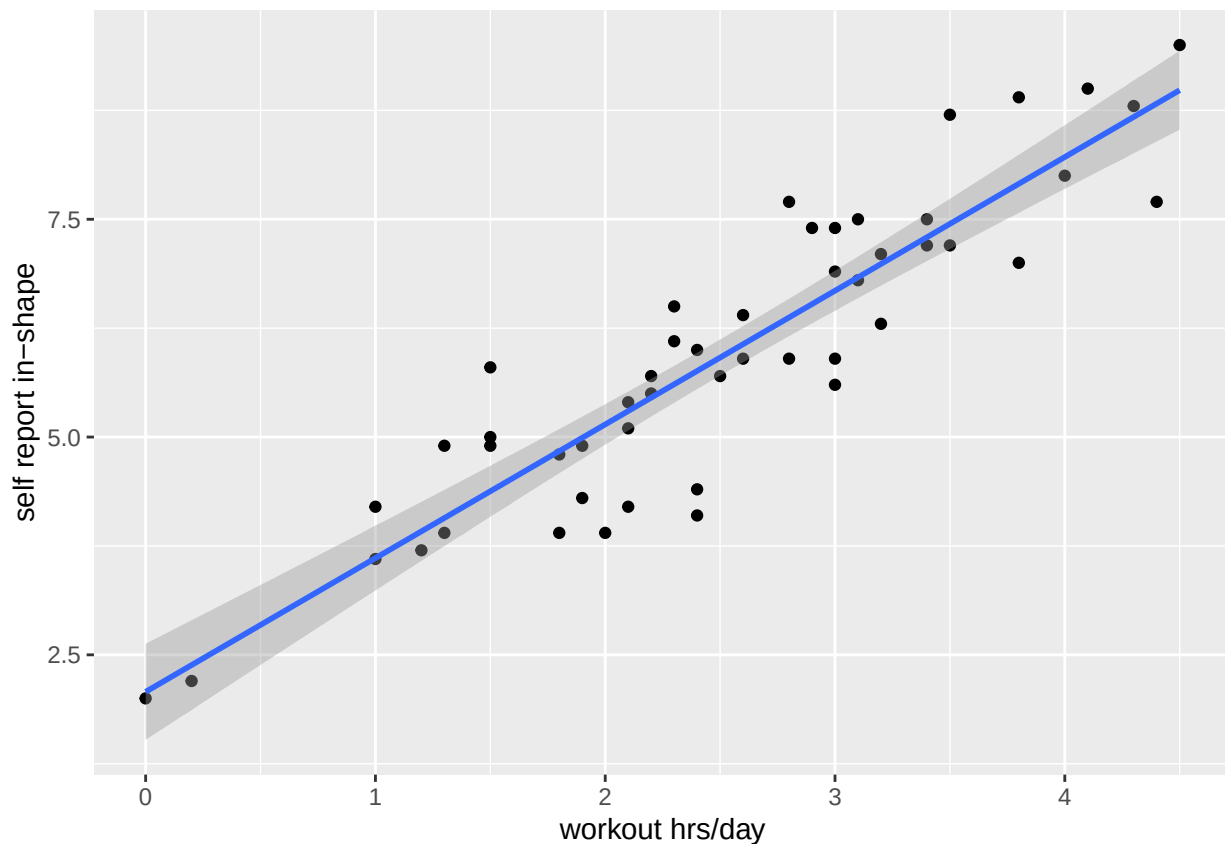
Load the first dataset

```
inShapeData <- read_csv('shapeVsHours.csv')
```

1.1 Fit the data with a straight line

```
plt_a <- inShapeData %>%
  ggplot(aes(x = hours, y = reported_shape)) +
  geom_point() +
  geom_smooth(method = 'lm') +
  xlab('workout hrs/day') +
  ylab('self report in-shape')
show(plt_a)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



1.2 Plot the residuals to check the fit

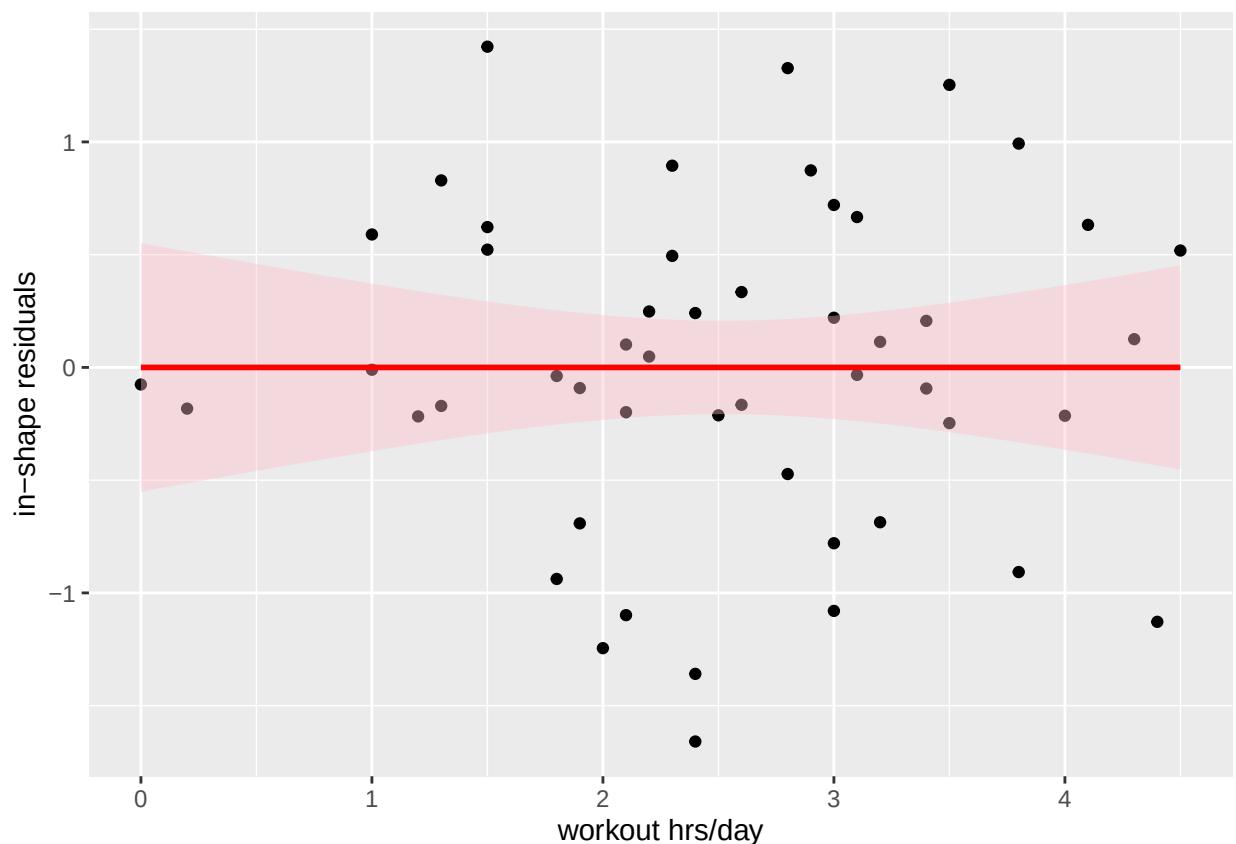
Fit the data with a straight line above in order to get the residuals.

```
a <- (lm(inShapeData$reported_shape ~
        poly(inShapeData$hours, 1)))
```

Now plot the residuals - the “leftovers” after the fit.

```
tib_a_res <- tibble(x = inShapeData$hours, y_res = a$residuals)
plt_a_res <- tib_a_res %>%
  ggplot(aes(x = x, y = y_res)) +
  geom_point() +
  geom_smooth(method = 'lm', color = 'red', fill = 'pink') +
  xlab('workout hrs/day') +
  ylab('in-shape residuals')
show(plt_a_res)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Note the lack of any pattern or trend in the residuals. This indicates that our straight-line fit has captured the essence of the data except for the inevitable random error.

Visually compare the above two plots, focusing on the left-most 10 data points to make sure you understand the relationship between these plots.

2. Load bpmVsHours.csv and take a look

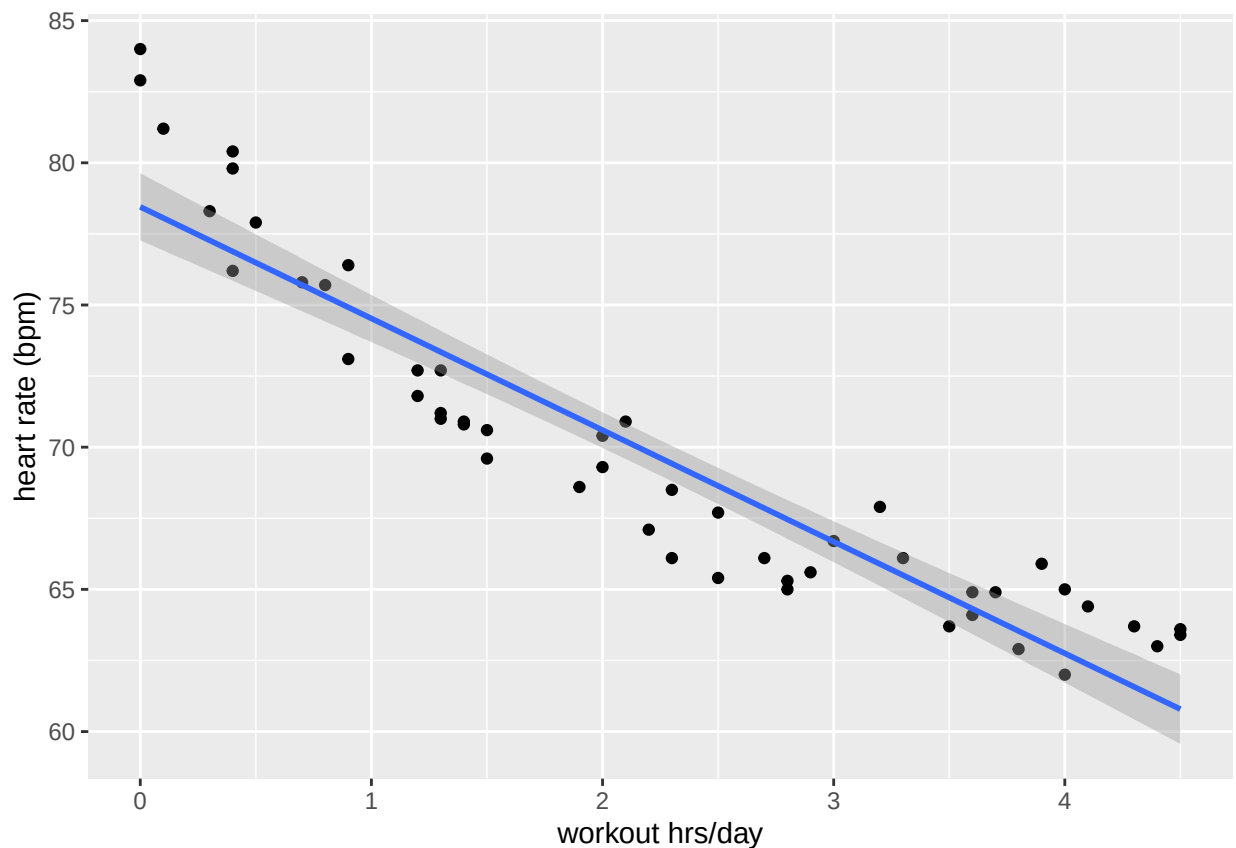
Load the heart rate data.

```
heartRateData <- read_csv('bpmVsHours.csv')
```

2.1 Fit the data with a straight line

```
plt_b <- heartRateData %>%
  ggplot(aes(x = hours, y = bpm)) +
  geom_point() +
  geom_smooth(method = 'lm') +
  xlab('workout hrs/day') +
  ylab('heart rate (bpm)')
show(plt_b)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Notice that this fit overestimates the data near the middle (the data points are mostly below the line), and underestimates the data at the ends (the data points are mostly above the line). This indicates that there is a curve in the data that the straight line fit is not capturing.

We can see this more clearly by looking at the residuals...

2.2 Plot the residuals to check the fit

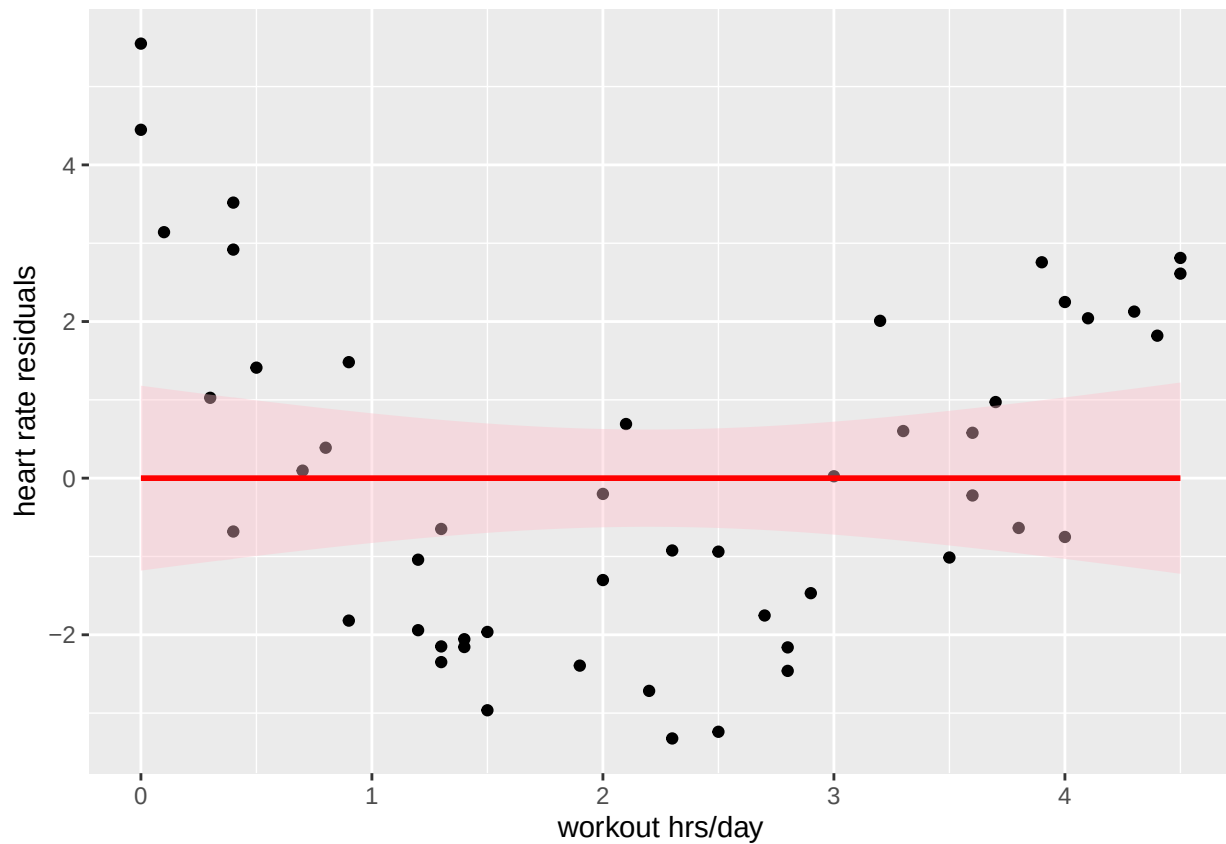
Fit the data with a straight line above in order to get the residuals.

```
b <- (lm(heartRateData$bpm ~
  poly(heartRateData$hours, 1)))
```

Let's look at the residuals.

```
tib_b_res <- tibble(x = heartRateData$hours, y_res = b$residuals)
plt_b_res <- tib_b_res %>%
  ggplot(aes(x = x, y = y_res)) +
  geom_point() +
  geom_smooth(method = 'lm', color = 'red', fill = 'pink') +
  xlab('workout hrs/day') +
  ylab('heart rate residuals')
show(plt_b_res)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

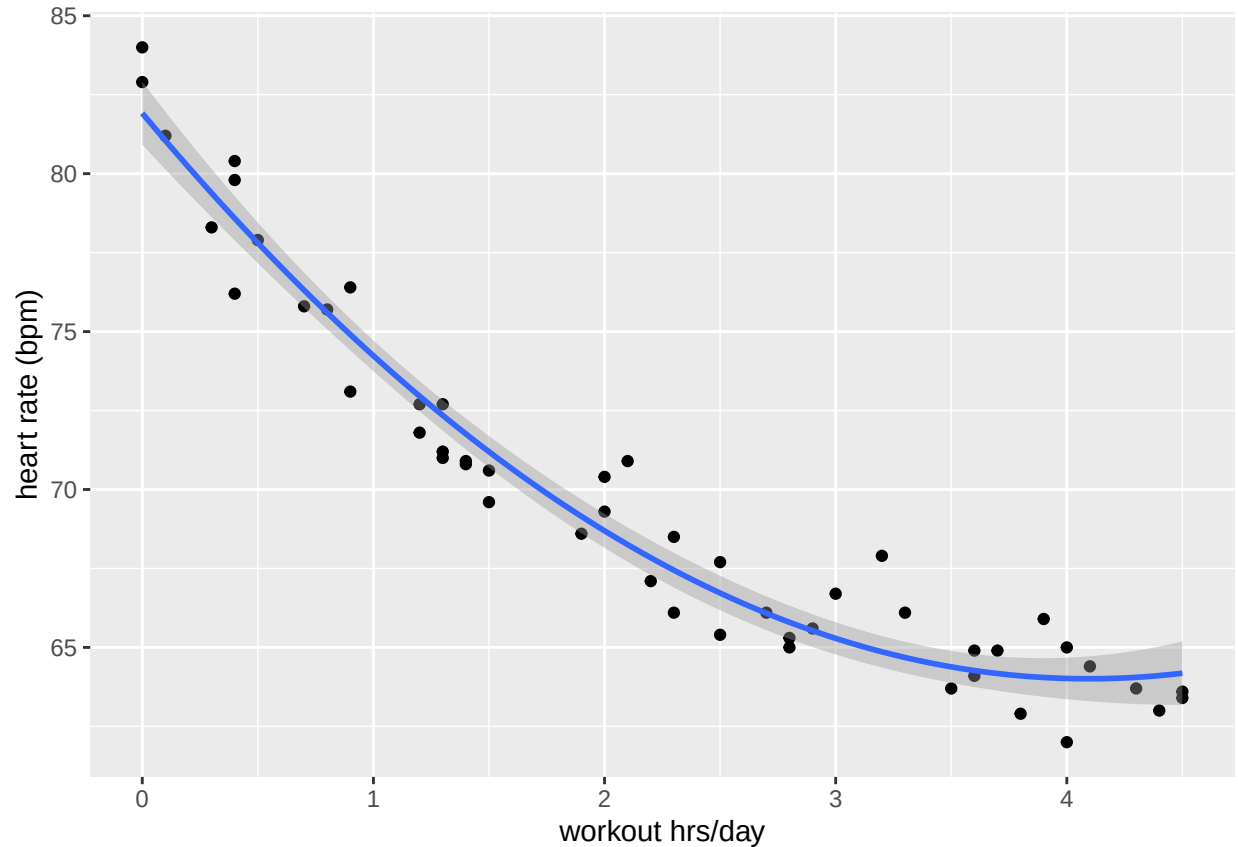


From this plot, we can clearly see the pattern of under- and overestimation. This fit is not doing it for us. So let's try a fit with a curve.

2.3 Fit the data with a curved line

Fit the data with a linear second order polynomial.

```
plt_b2 <- heartRateData %>%
  ggplot(aes(x = hours, y = bpm)) +
  geom_point() +
  geom_smooth(method = 'lm', formula = y ~ poly(x, 2)) +
  xlab('workout hrs/day') +
  ylab('heart rate (bpm)')
show(plt_b2)
```



That looks much better! Now let's check the residuals.

2.4 Plot the residuals for the second order fit

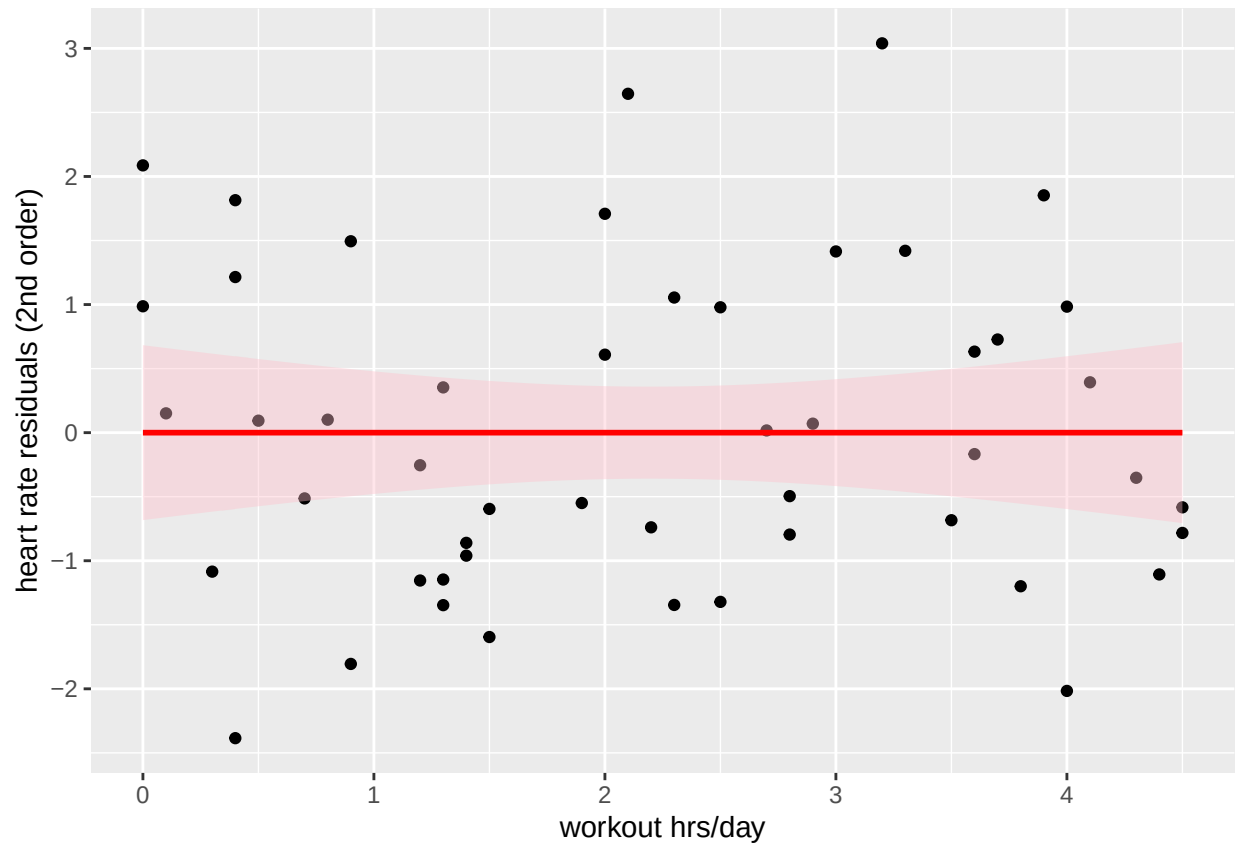
Fit the data with a curved line above in order to get the residuals.

```
b2 <- (lm(heartRateData$bpm ~
          poly(heartRateData$hours, 2)))
```

Plot the residuals.

```
# plot residuals for 2nd order fit
tib_b2_res <- tibble(x = heartRateData$hours, y_res = b2$residuals)
plt_b2_res <- tib_b2_res %>%
  ggplot(aes(x = x, y = y_res)) +
  geom_point() +
  geom_smooth(method = 'lm', color = 'red', fill = 'pink') +
  xlab('workout hrs/day') +
  ylab('heart rate residuals (2nd order)')
show(plt_b2_res)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Great! Now the residuals look random, indicating that our fit has accounted for the systematic trend in the data, and only random error remains.

Deliverables: Productivity dataset

3. Load prodVsHours.csv and take a look

3.1 Fit the data with a straight line

Plot the productivity data with a straight line (first order polynomial)

Plot summary:

3.2 Plot the residuals to check the fit

Do the regression.

Plot the residuals from the regression to check the fit.

Plot summary:

3.3 Fit the data with a curved line

Plot the data with second order linear fit (a curved line).

3.4 Plot the residuals for the second order fit

Do the regression.

Plot the residuals from the regression to check the fit.

Plot summary:

4. Big picture summary

Write a few sentences summarizing the big picture. How does hrs/week of workout affect self-perceived shape, heart rate, and perceived productivity? Are all the data described by straight lines? Do all data necessarily increase or decrease overall? What information do residuals give you?